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Q1- One amu is equal to:

(a) 1.66×10^{-24} g

(b) 1.66×10^{-24} kg

(c) 6.02×10^{23} g

(d) None of these

Q2- The mass of one molecule of HNO_3 is:

(a) $63 \times 1.66 \times 10^{-24}$

(b) 63 amu

(c) $63 \times 1.66 \times 10^{-27}$ kg

(d) all of these

Q3- The number of O atoms in 98 amu of H_2SO_4 is:

(a) 1 mole

(b) 4 moles

(c) 4

(d) 1

Q4- 6×10^{22} atoms of an element weight 13.8 g. The atomic mass of the element is:

(a) 120 a.m.u.

(b) 35.5 a.m.u.

(c) 108 a.m.u.

(d) 180 a.m.u.

Q5- Number of electrons in 168 amu of 36/25 Fe^{2+} ions will be:

(a) 75 N_A

(b) 78

(c) 72

(d) 72 N_A

Q6- The number of neutrons present 124 amu of P_4 is (Given: P):

(a) 16

(b) 16 N_A

(c) 64 N

(d) 64

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Q7- Calculate number of neutrons present in 12×10^{25} atoms of oxygen ($_8O^{17}$) : (Given: $N_A = 6 \times 10^{23}$)

(a) 1800

(b) 1600

(c) 1800 N_A

(d) 3200 N_A

Q8- If mass of one atom is 3.32×10^{-23} g then calculate number of nucleons (neutrons and protons) present in 2 atoms of the element:

(a) 40

(b) 20

(c) 10

(d) 40 N_A

Q9- What is the number of moles of O-atom in 126 amu of HNO_3 ?

(a) 2

(b) $2/N_A$

(c) 6

(d) $6/N_A$

Q10- A sample of sodium has a mass of 46 g. What is the mass of the same number of calcium atoms as sodium atoms present in given sample ?

(a) 46 g

(b) 20 g

(c) 40 g

(d) 80 g

Q11- What is the charge of 96 amu of S?

(a) 2C

(b) 3.2×10^{-19} C

(c) 9.6×10^{-19} C

(d) 6C

Q12- Calculate number of electrons present in 9.5 g of PO_4^{3-}

(a) 6

(b) $5N_A$

(c) $0.1N_A$

(d) 4.7 N_A

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Q13- An atom of an element is 3.5 times heavier as compared to one atom of ^{12}C isotope. The molar mass of element (g / mol) is:

(a) 3.5

(b) 12

(c) 42

(d) None of these

Q14- Number of electrons in 28 g of Fe ions will be:

(a) 12

(b) 0.5

(c) 12 N_A

(d) 0.5 N_A

Q15- The total no. of neutrons present in 54 mL. $H_2O_{17}$are:

(a) $3N_A$

(b) 30 N_A

(c) 24 N

(d) none of these

Q16- Total no, of electrons present in 48 g Mgare:

(a) 24 N_A

(b) $2N_A$

(c) $20N_A$

(d) none of these

Q17- The number of neutron in 5 g of D_2O (D is $^{2}_{1}H$)

(a) 0.25 N_A

(b) 2-5 N_A

(c) 1-1N_A

(d) none of these

Q18- 3.011 x 10^{21} atoms of an element weight 1.15 gm. The atomic mass of the element is:

(a) 10 amu

(b) 2.3 amu

(c) 35.5 amu

(d) 23 amu

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Q19- 10 mole of CO_2 contains:

(a) 220 electrons

(b) 440 mole protons

(c) 220 mole neutrons

(d) 220 x 6.02 x 10^{23} nucleons

Q20- Mass of oxygen present in 750 amu sample of $CuSO_4$,SH_2O : (Given Cu = 64)

(a) 64 amu

(b) 144 amu

(c) 80 amu

(d) None of these

Q21- How many moles of $Ca_3(PO_4)_2$ will contain 0.20 mol of oxygen atoms?

(a) 0.25

(b) 0.025

(c) 40

(d) None of these

Q22- How many moles of proton weight one mg? (Weight of one proton = 1.66 x 10^{-24} gm)

(a) 10^{-3}

(b) 6.022 x 10^{-3}

(c) 6.022 x 10^{3}

(d) 1.66 x 10^{-3}

Q23- How many hydrogen atoms are present in 0.1 mole of mohr's salt ($FeSO_4$ $(NH_4)_2$ SO_4 . $6H_2O)_2$

(a) 10 × N_A

(b) 14 × N_A

(c) 2 × N_A

(d) None of these

Q24- The number of electrons present in 3.2 g-ion of O^{3-} is:

(a) 32N_A

(b) 3.2 N_A

(c) N_A

(d) 2.0 N_A

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Q25- Cisplatin, an anticancer drug, has the molecular formula Pt $(NH_3)_2Cl_2$ What the mass (in gram) of one molecule? (Atomic weights: p_t= 195 H = 1 N = 14, Cl = 35.5)

(a) 4.98×10^{-21}

(b) 4.98×10^{-25}

(c) 6.55×10^{-21}

(d) 3.85×10^{-12}

Q26- Aspirin has the formula C_9H_5O4 How many atoms of oxygen are 360 mg ^{2} there in a tablet weighing

(a) 1.204×10^{23}

(b) 1.08×10^{22}

(c) 1.204×10^{24}

(d) 4.81×10^{2}

Q27- 20 g of an ideal gas contains only atoms of S and O occupies 5.6 L at 1 atm and 273 K. What is the mol. wt. of gas ?

(a) 64

(b) 80

(c) 96

(d) None of these

Q28- A sample of ammonium phosphate, $(NH_4)_3PO_4$ contains 6 moles of hydrogen atoms. The number of moles of oxygen atoms in the sample is:

(a) 1

(b) 2

(c) 4

(d) 6

Q29- Total number of moles of oxygen atoms in 3 litre O_3(g) at 27 ºC and 8.21 atm are:

(a) 3

(b) 1

(c) 1

(d) none of these

Q30- One atom of an elementxweigh 6.643×10^{-23} . Number of moles of atom in 20 kg is:

(a) 4

(b) 40

(c) 100

(d) 500

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Q31- Mass of one atom of the element A is 7.0854×10^{-23} of the element A? g. How many atoms are contained in 1 g

(a) 2.509×10^{23}

(b) 6.022×10^{21}

(c) 12.044×10^{12}

(d) None of these

Q32- Which of the following contains the largest mass of hydrogen

(a) 5.0 moles $C_2H_2O_4$

(b) 1.1 moles $C_3H_4O_3$

(c) 1.5 moles $C_6H_6O_6$

(d) 4.0 moles $C_2H_4O_2$